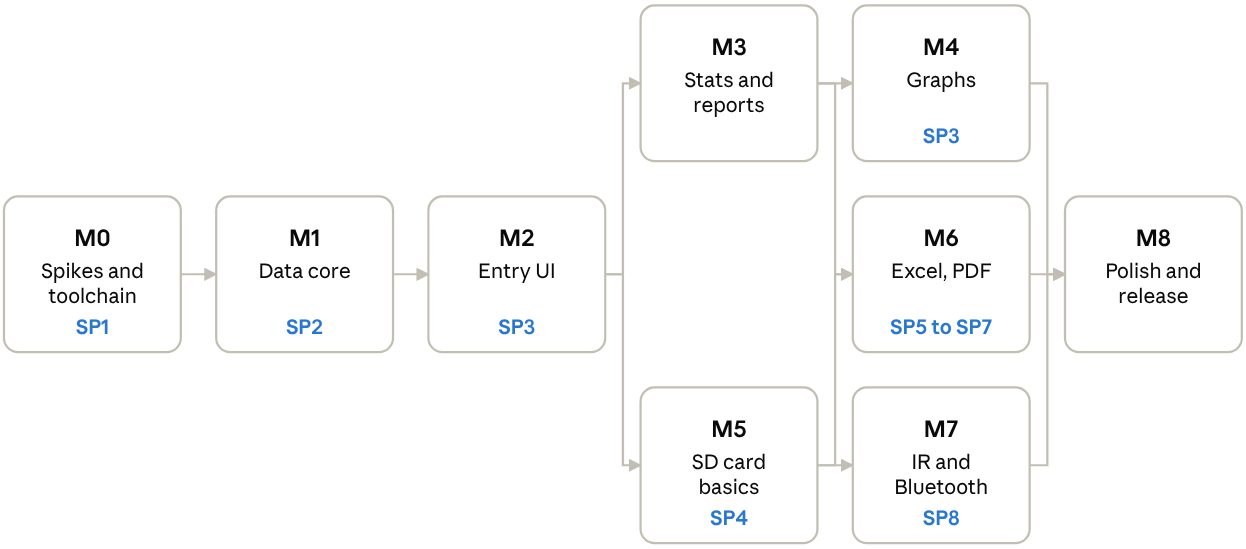
Backlog

This is the build plan for FuelLog: nine milestones, eight spikes and 36 user stories. Spikes run first so that no milestone starts on an unproven assumption.

Roadmap

Two tracks split after Entry UI and rejoin at release

Milestones M0 to M8; coloured tags name the spike that unblocks each one



roadmap · 9 milestones, 8 spikes

After M2 the work runs in two tracks: reports and graphs on top, card and transfer below. They join at M6, which needs both report output and the SD writers, and everything meets at M8.

Milestones

Each milestone ends with a .prc that installs and runs on the T|X, plus green host tests. A milestone is done only when its exit criteria hold on the device, not just in the emulator.

ID	Milestone	Goal	Exit criteria	Needs	Stories
M0	Spikes and toolchain	Prove the risky assumptions	SP1 to SP8 closed with written answers; hello-world .prc runs on the T|X; `make test` runs on macOS; hi-res mock-ups of every form approved (see the Screen designs tab); launcher icon pixels hand-tuned from the reference	none	none
M1	Data core	Databases, maths and settings everything else sits on	three databases create, open and upgrade; calc library passes host tests for all units, currencies and EV; Prefs persist	M0	US-01 to US-03
M2	Entry UI	Log real fill-ups daily	a fill is logged in under 20 seconds; vehicles, programmes, currencies, payment and charging all editable on device	M1	US-04 to US-12
M3	Stats, reports and service	Answer "how am I doing" on the device, and what service is due	stats match a hand-checked spreadsheet of 50 fills to the cent and to 0.1 L/100 km; service due dates match a hand-worked example for distance, time and projected pace; the alarm fires after a soft reset	M2	US-13 to US-16, US-33 to US-36
M4	Graphs	See trends at a glance	three graphs draw at double density in under 1 s for 500 fills	M3	US-17, US-18
M5	SD card basics	Get data out, bring paper logs in, never lose it	CSV round-trips; paper-log import works; backup and restore survive a hard reset	M2	US-19 to US-22
M6	Excel and PDF	Office-grade outputs and Excel import	XLSX opens clean in Excel and Numbers; PDF opens in Preview and Acrobat; XLSX and XLS import pass fixtures	M5, M3	US-23 to US-26

ID	Milestone	Goal	Exit criteria	Needs	Stories
M7	IR and Bluetooth	Move data between Palms and off the device	T|X to LifeDrive merge with no duplicates; an export sent over Bluetooth lands on the other Palm's SD card	M5	US-27 to US-29
M8	Polish and release	Feels native, runs everywhere it should	DIA and 5-way on every form; LifeDrive pass; Go PDB reader ships; v1.0 tagged	M4, M6, M7	US-30 to US-32

Spikes

Each spike is timeboxed throwaway code that ends in a written answer, not production code. All eight fit in about six working days, and a spike that fails its timebox triggers its fallback instead of eating into a milestone.

ID	Question	Timebox	Done when	Fallback	Unblocks	Status
SP1	Does the jichu4n Homebrew tap build prc-tools on Apple Silicon macOS 27?	1 day	hello-world .prc built by `make`, installed on the T|X	`linux/amd64` Docker image with the same tools	M0	To do
SP2	Can GCC 2.95 `long long` and 32-bit fixed point hold lifetime totals without overflow?	0.5 day	worst-case bounds written down (20 years, 4 vehicles, 16 programmes) and host-tested	64-bit only in sums; per-year subtotals	M1	To do
SP3	Hi-res and DIA on the T|X: a form that resizes when the input area collapses, drawing at double density	1 day	one sample form plus a line graph working both ways	fixed 320x320 layout with a scrolling table	M2, M4	To do
SP4	VFS on the T|X: write speed, folder creation, `VFSExportDatabaseToFile` round trip	0.5 day	200 KB written and timed; exported .pdb re-imported intact	write in smaller chunks; backup via HotSync only	M5	To do

ID	Question	Timebox	Done when	Fallback	Unblocks	Status
SP5	Does a stored (uncompressed) zip XLSX made by our writer open clean in Excel and Numbers?	1 day	host-generated file opens with no repair prompt in both	add sharedStrings or switch to SpreadsheetML 2003 XML	M6	To do
SP6	Inflate on 68k under PACE: speed and heap for a 200 KB .xlsx	1 day	timed decode of a Numbers-saved file on the T|X	decompress to card in passes; or XLSX import via desktop CSV	M6	To do
SP7	Base-14 PDF from our writer: renders in Preview, Acrobat and iOS?	0.5 day	a 3-page sample opens in all three with correct euro sign	embed nothing, drop the euro glyph to "EUR"	M6	To do
SP8	Exchange Manager Palm to Palm: does `_btobex:` from the T|X reach the LifeDrive, and does a `.fuel` envelope receive with the app closed?	0.5 day	a 200 KB export crosses by Bluetooth and IR; the receiver writes it to SD; a record `.fuel` opens the import dialog	IR only for v1; move files by SD card swap	M7	To do

User stories

There are 36 stories across M1 to M8, each small enough to finish inside its milestone. Sizes are S (half a day or less), M (one to two days) and L (three to five days). An L that grows past five days is split before it starts.

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-01	M1	have my log stored in metric fixed point	switching units never changes my data	stored bytes identical after every unit switch; conversions round-trip within 0.1 of the display unit	M	To do

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-02	M1	pick distance, volume, economy and price units separately, with an odometer unit per vehicle	the app reads the way I think and the way each car's dash reads	every screen, report and export follows Prefs; a miles-odometer car is entered in miles	M	To do
US-03	M1	set a home currency and keep a rate per currency	trips across the border still roll up into one total	USD, CAD, EUR and AUD selectable; default currency for new entries set in Prefs (fixed or last used) with a per-vehicle override; fxRate stored per fill; totals unchanged when rates are edited later	M	To do
US-04	M2	log a fill-up from New in a few taps	logging at the pump is quicker than paper	date, vehicle, payment and grade pre-filled; any two of volume, price and total compute the third; under 20 s end to end	L	To do
US-05	M2	edit or delete a past fill	mistakes don't poison my stats	delete asks first; balances and economy recalculate at once	S	To do
US-06	M2	mark a fill as full tank or after a missed fill	economy stays honest	missed fill breaks the chain; partial fills roll into the next full; trip meter reading cross-checks the odometer and warns on a gap over 2%; blank odometer derived from trip	S	To do
US-07	M2	record points for several programmes on one fill	linked programmes such as Petro-Points and Triangle Rewards both count	up to 4 programmes per fill; auto-apply pre-fills by earn rule and payment; every line editable; per-gallon or per-litre redemptions (Kroger, PC Optimum) compute the discount within the volume and per-unit caps	M	To do

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-08	M2	add, rename and archive vehicles with powertrain, tank or battery, and odometer unit	the whole household fleet lives in one log	up to 16 vehicles; delete blocked while fills exist; archived cars hidden but reportable	M	To do
US-09	M2	manage rewards programmes, starting from the presets	I track the programmes I actually use	6 presets (incl. Kroger Fuel Points for Fred Meyer and King Soopers) ship with known default earn, redemption and expiry rules; add, edit, archive; money-valued programmes show dollars	M	To do
US-10	M2	log a charging session for an EV or plug-in hybrid	electricity costs sit beside fuel	kWh, price per kWh (0 allowed), home, public or DC fast, charged-to-full flag; home rate pre-fills	M	To do
US-11	M2	record Cash or Card for each fill	I can see how I pay and card points only apply to card fills	push group defaults to last used; programmes with `requiresPayment` follow it	S	To do
US-12	M2	enter a fill in its receipt currency	receipts match the log exactly	currency popup on the form; rate pre-filled from the table and editable	S	To do
US-13	M3	see lifetime stats for one vehicle or all on one screen	I know my numbers at a glance	the Stats table from the spec fits one hi-res screen; matches the hand-checked fixture	M	To do
US-14	M3	run monthly, yearly and custom-range reports	I can compare periods	period popup; 5-way steps through periods; per-vehicle breakdown under All vehicles	M	To do
US-15	M3	see each programme's balance and its value	I know what my points are worth	balance = opening + earned - redeemed per programme; value shown when set; adjustments counted; savings from redemptions totalled; points expiring this month flagged	S	To do

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-16	M3	see spend split by currency and by payment method	cross-border and cash spending is visible	report sections for each; totals reconcile to the home-currency total	S	To do
US-17	M4	graph economy over time and tap a point for its value	I spot trends and outliers	full-to-full segments plotted; min, max and mean labelled; tap shows date and value	L	To do
US-18	M4	graph price per fill and spend per month, one line per vehicle	I see where the money goes	popup switches graph; All vehicles draws a line per car; under 1 s for 500 fills	M	To do
US-19	M5	export the log to CSV on the SD card	I can open it in any spreadsheet	file in `Export/`; UTF-8 with BOM; header with units; respects filters	S	To do
US-20	M5	import my filled paper logs from CSV	years of history come with me	preview with errors and line numbers before writing; units read from header; re-import adds no duplicates	L	To do
US-21	M5	back up to SD on demand and automatically	I never lose the log	dated set of 3 PDBs; keeps last 5; auto after 7 days on exit; HotSync backup bit set	M	To do
US-22	M5	restore from a backup set	a lost or reset Palm is recoverable	lists sets; confirm before replacing; newer schema refused; older schema migrated	M	To do
US-23	M6	export to XLSX with Fills, Monthly, Vehicles and Programmes sheets	I get a real workbook, not text	numbers and dates are typed cells; currency formats applied; opens with no repair prompt	L	To do
US-24	M6	export a PDF report laid out like the paper log	I can print or file it	summary page, log pages, economy graph; Letter landscape; euro sign renders	L	To do
US-25	M6	import an XLSX workbook	spreadsheets from Excel or Numbers load directly	reads Fills or first sheet; 1900 and 1904 date systems; own export round-trips exactly	L	To do

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-26	M6	import an old XLS workbook	older files need no conversion	BIFF8 cells and cached formula values read; same preview and merge as CSV	L	To do
US-27	M7	beam a fill, vehicle or programme over IR	sharing with another Palm is instant	`.fuel` received with app closed; merge offered; duplicate beam is a no-op	M	To do
US-28	M7	send my whole log to my other Palm and merge	the T|X and LifeDrive agree	all three databases in one bundle; Merge or Replace; ids remapped on clash	M	To do
US-29	M7	send an export over Bluetooth to another Palm	the other Palm gets the report without swapping cards	CSV, XLSX or PDF sent in a `.fuel` envelope; the receiver writes it to its `Export/` folder with the original name; CSV and XLSX offer import	S	To do
US-30	M8	use the full 320x480 screen and the 5-way everywhere	it feels like a native T|X app	every form resizes on DIA collapse; 5-way moves and selects on all lists and forms	M	To do
US-31	M8	run FuelLog on the LifeDrive and older 68k Palms	my other devices work too	full pass on LifeDrive; launches low-res in CloudpilotEmu without crashing	S	To do
US-32	M8	read FuelLog backups on my Mac with a Go tool	my data is never locked in the Palm	`fuellog dump` prints fills as CSV from the three PDBs; matches device export	M	To do

Service tracking (added to M3; it needs the M2 entry UI and the fill odometer)

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-33	M3	set a maximum distance and time between oil changes, and other services, per vehicle	each car follows its own manual	presets by powertrain; intervals blank until set; warn-ahead defaults 10% and 14 days; first-use asks for the last service date and odometer	M	To do

ID	M	As a driver, I want to…	So that…	Acceptance criteria	Size	Status
US-34	M3	log a service visit covering several items at once	one shop visit is one entry	checklist of service types plus Other with free text (up to 24 characters, promotable to a tracked type); odometer pre-filled from latest fill; cost, currency, Cash/Card, shop, detail	M	To do
US-35	M3	see what is due soon or overdue, and be reminded when FuelLog is closed	I never run past an oil change	Service list shows km and days left and projected date at 90-day pace; wrench on fill list; one alert when a fill tips into Due soon; `AlmSetAlarm` set and restored after reset	L	To do
US-36	M3	see service history, the longest gap I actually reached, and running cost	I know how I really maintain each car	history per service with over-interval gaps flagged; running cost per 100 km (fuel plus service); Services sheet, CSV and PDF page in exports	M	To do

Definition of done

A story is done only when every box below holds on the real T|X. That is how polish is enforced throughout the build instead of being left to M8.

- [ ] Acceptance criteria met on device, not only in CloudpilotEmu
- [ ] Form matches its approved hi-res mock-up in the Screen designs tab; aligned to the grid; numbers right-aligned on the decimal
- [ ] Resizes cleanly when the input area collapses and opens; nothing clipped in the 320x320 fallback
- [ ] 5-way reaches every control with a visible focus ring; centre selects
- [ ] Colours come from the device Colour Theme; status never shown by colour alone
- [ ] No flicker on scroll or redraw; launch and save inside the speed targets
- [ ] Every error path shows a plain-words alert that says what to do next
- [ ] Empty state written; Tips (?) text written

- [ ] Host unit tests green; no new compiler warnings
- [ ] State restored after switching apps, and after an interruption mid-entry